

Generating data and outlining algorithms

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Generating the data

Let $\mathbf{X} \in \mathbb{R}^n$ be given by

$$\mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} k \\ 2k \\ \vdots \\ nk \end{bmatrix}$$

with $k \in \mathbb{R}^+$, $n \in \mathbb{N}$.

Let $\mathbf{S} \in \mathbb{R}^p$ be given by

$$\mathbf{S} = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_p \end{bmatrix}$$

with $p \in \mathbb{N}$, representing the location of steps, with $s_1 < s_2 < \dots < s_p$, where s_i is the location of a step, with $s_p < n$.

Let $\mathbf{T} \in \mathbb{R}^{p+1}$ be given by

$$\mathbf{T} = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_{p+1} \end{bmatrix}$$

representing additional shift values for each step.

Let L be a function defined by

$$L_{\mathbf{S}, \mathbf{T}}(x_i) = \begin{cases} \log\left(\frac{x_i}{x_1}\right) + t_1, & x_i \leq x_1 s_1, \\ \log\left(\frac{s_1(x_i - x_1 s_1)}{x_1}\right) + t_2, & x_1 s_1 < x_i \leq x_1 s_2, \\ \log\left(\frac{s_1 s_2(x_i - x_1 s_2)}{x_1}\right) + t_3, & x_1 s_2 < x_i \leq x_1 s_3, \\ \vdots & \\ \log\left(\frac{(\prod_{i=1}^{p-1} s_i)(x_i - x_1 s_{p-1})}{x_1}\right) + t_p, & x_1 s_{p-1} < x_i \leq x_1 s_p, \\ \log\left(\frac{(\prod_{i=1}^p s_i)(x_i - x_1 s_p)}{x_1}\right) + t_{p+1}, & x_i > x_1 s_p, \end{cases}$$

Let $\boldsymbol{\epsilon} \in \mathbb{R}^n$ be a continuous multivariate random variable

$$\boldsymbol{\epsilon} \sim F(\boldsymbol{\theta})$$

Finally, let the vector $\mathbf{Y}$ be defined by

$$\mathbf{Y} = L_{\mathbf{S}, \mathbf{T}}(\mathbf{X}) + \boldsymbol{\epsilon},$$

where $L_{\mathbf{S}, \mathbf{T}}(\mathbf{X}) \in \mathbb{R}^n$ denotes $L_{\mathbf{S}, \mathbf{T}}$ applied element-wise to $\mathbf{X}$

Then the data is given by $(\mathbf{X}, \mathbf{Y})$

For the methods described in the next section, the following parameters are used

$$k = 0.01$$

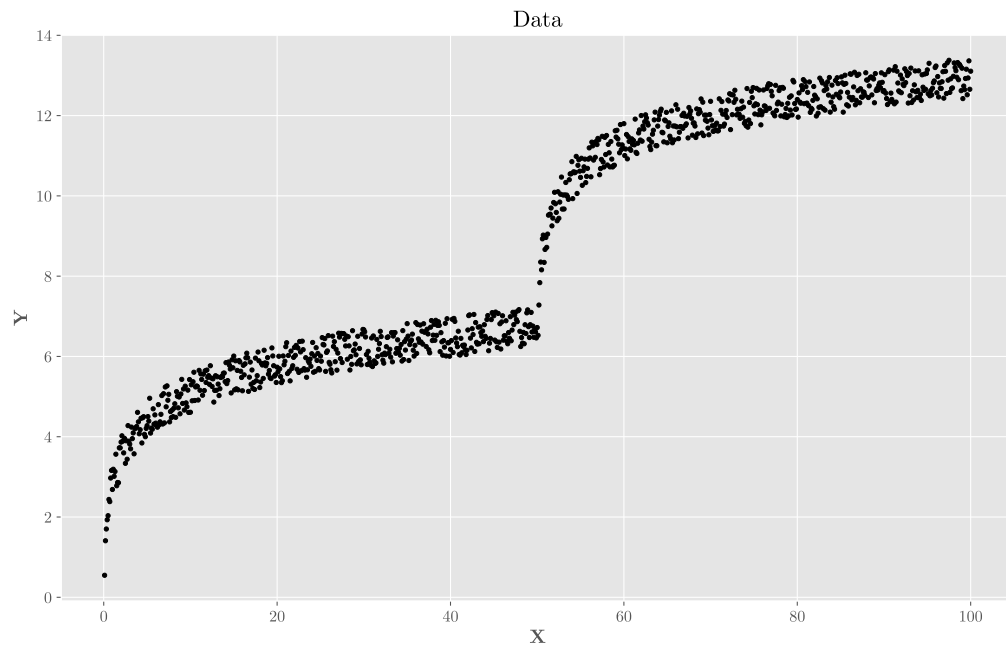
$$n = 1000$$

$$\mathbf{S} = [500]$$

$$\mathbf{T} = [0]$$

$$\mathbf{F} = \text{Unif}(0, 1)$$

These parameters generate the data shown below



Methods

Interval gradient method

In this method, the data is split up into intervals of width w . Within each interval, a linear model is fitted, and the slope is recorded. If a slope exceeds a threshold λ , a step is detected.

This method works because the data gets gradually flatter as x increases, but when there is a step, the y value will increase quickly, meaning the slope of a linear model will be much steeper there.

Through testing, an accurate value for λ is given by $\frac{1.5}{wk}$

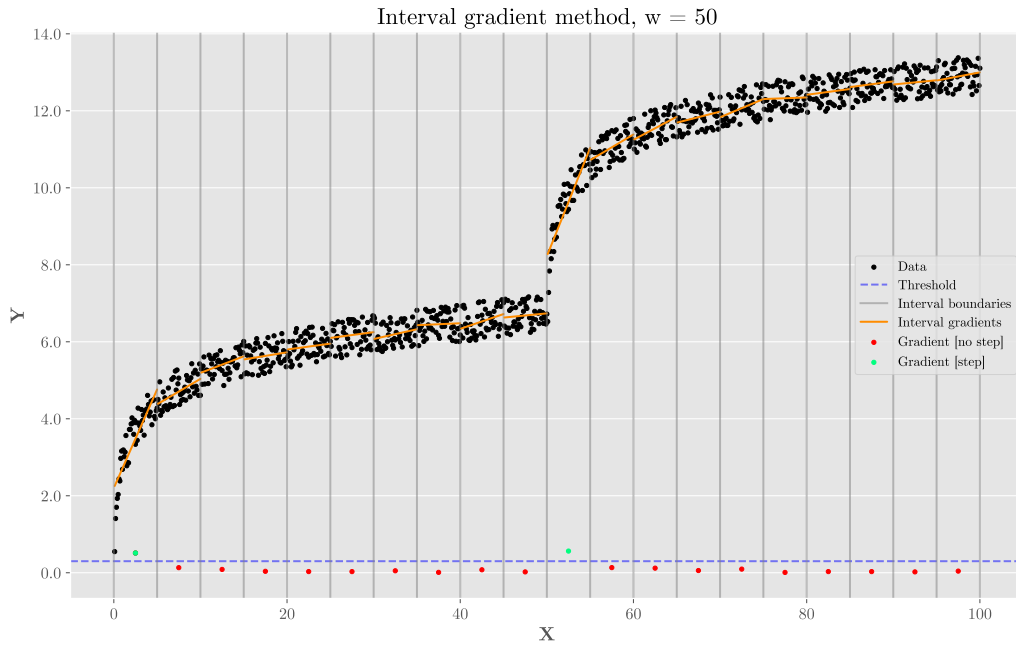


Figure 1: Interval gradient method with interval width $w = 50$

Total gradient method

This method also splits the data up into intervals of width w . Instead of gradients being calculated for each interval, gradients are calculated in intervals of increasing number. The slope is recorded, and a slope that is a local minimum and is less than neighbouring slopes by a threshold γ is detected as a step.

This works because a linear model fitted to log data will have a decreasing slope until the data jumps, after which it will jump up and begin to decrease. So the data will step within the interval where the slope is smaller than the neighbouring slopes. To account for falsely detected step due to noise, the slope difference between the local minimum and neighbouring steps must exceed a certain threshold.

Through testing, an accurate value for γ is 0.001

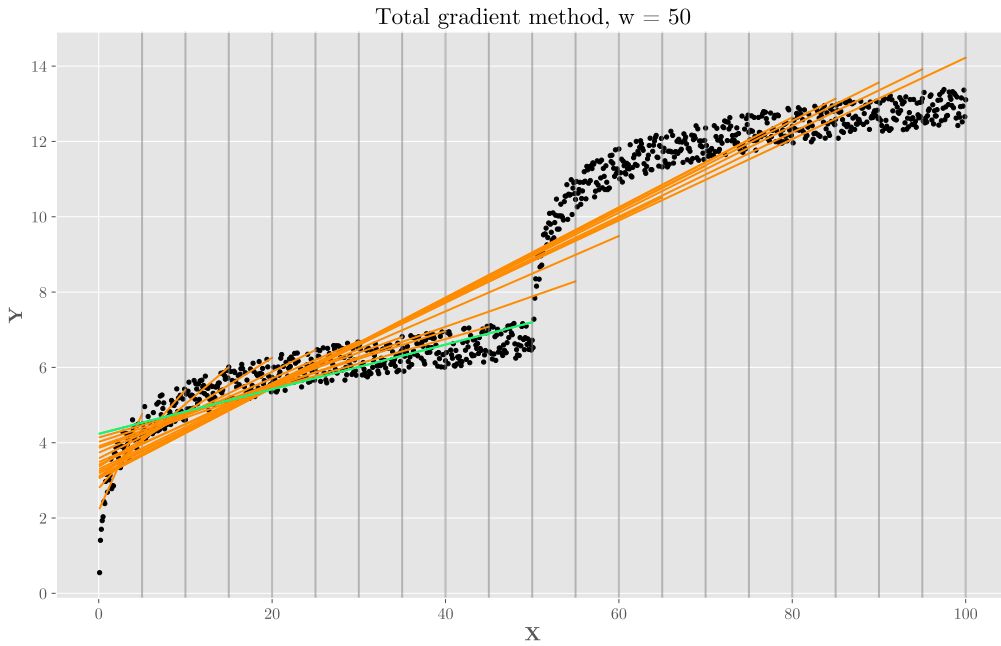


Figure 2: Total gradient method with interval width $w = 50$

KDE method

In this method, a kernel density estimate is calculated for the y values. The number of steps is equal to 1 less than the number of local maxima in the KDE. A Gaussian kernel is used, with $h = 0.5$

This method works because the y values are denser at the end of a logarithmic “section”. If there is one step, there will be 2 of these sections. So if there are 2 highly dense regions, there will be 1 step.

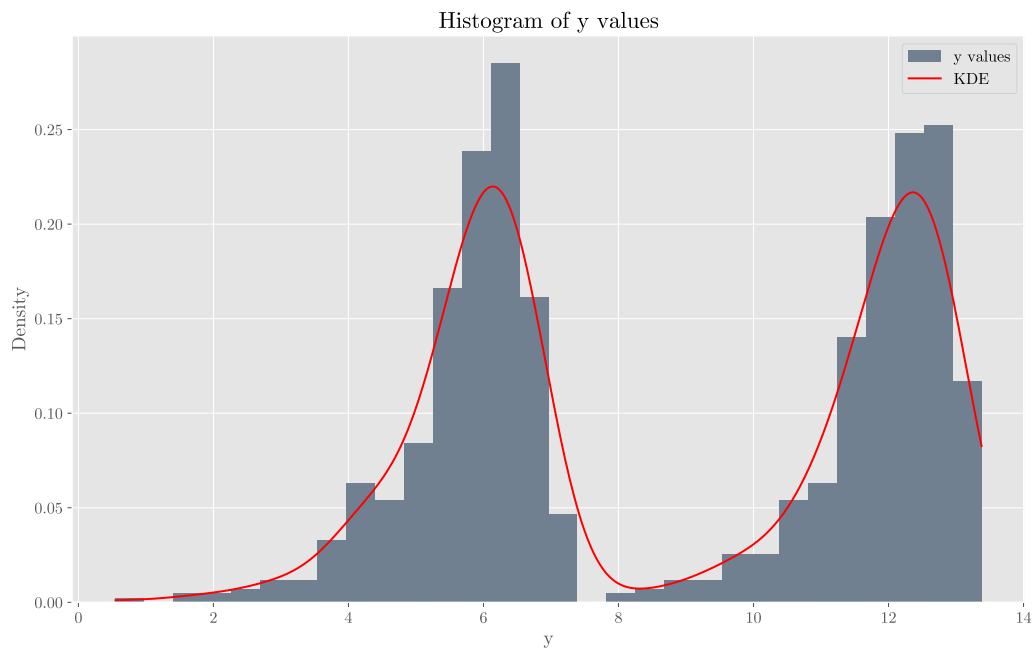


Figure 3: KDE method, Gaussian kernel, $h = 0.5$

Progressive F method

Similarly to the total gradient method, a model is fitted for progressively larger amounts of data. In this method, a logarithmic model is fitted, and the f statistic is recorded. The f statistics are smoothed, and a local maxima in the smoothed values means a step has been detected.

This method works because the f statistic of a model that fits the data will be higher. As soon as that model stops fitting the data as well, the f statistic will be lower. Therefore a local maximum will mean a step is present.

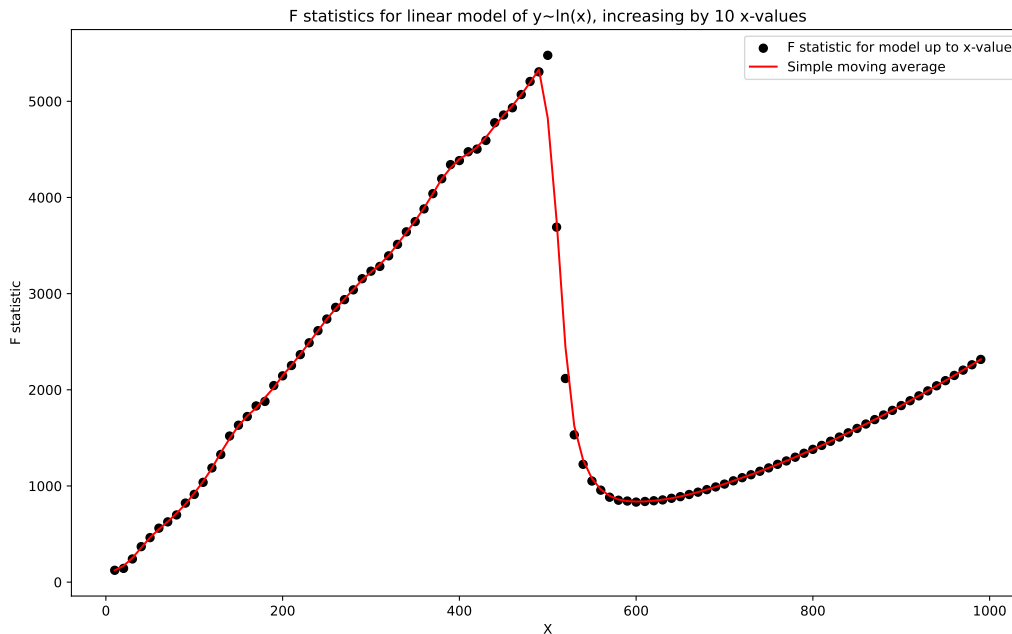


Figure 4: F statistic method with step = 10